Priced In? Stock-Market Reactions to CHIPS Act Award Announcements

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July 2026

Abstract

The 2022 CHIPS and Science Act directed tens of billions of dollars in manufacturing subsidies to semiconductor firms, disbursed through a sequence of company-specific award announcements in 2024. We ask whether these announcements moved the awardees' stock prices, and whether the reaction scaled with award size. Using a market-model event study on 15 publicly-traded U.S. awardees, we find *no* statistically significant average abnormal return around the first ("preliminary terms") announcement: the mean cumulative abnormal return over $[-1, +1]$ is $+2.6\%$ ($t = 0.99$), the median is $+0.5\%$, only 8 of 15 reactions are positive, and a pre-event placebo window is if anything negative. The null is invariant to the abnormal-return model (market

*Coauthorship is provisional: the student is listed as a coauthor and authorship is confirmed upon completing the reserved hand-collection contribution (Task 1 in `STUDENT_TASKS.md`). A note on author order and status: the student is listed first here as a program convention, but in finance and economics published author order is conventionally alphabetical; all authorship on this draft is tentative and specific to the ASSIP 2026 program at this stage.

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model, market-adjusted, mean-adjusted) and to a battery of standardized and non-parametric test statistics (Patell, Boehmer–Musumeci–Poulsen, generalized sign). A naive cross-sectional regression suggests the announcement return rises steeply with the award's size relative to firm market capitalization (OLS slope $t = 32$, $R^2 = 0.80$), but this is entirely an artifact of a single distressed micro-cap, Wolfspeed, whose $750M award equalled 52% of its market value: dropping it collapses the slope to $t = 1.0$ ($R^2 = 0.04$), and the outlier-robust Spearman rank correlation is only 0.37 ($p = 0.18$). Two additions push past the usual "we found nothing" event study. First, *randomization inference*—2,000 placebo event dates and 2,000 permutations of the award-size vector—confirms both halves of the story from a design-based standpoint: the mean CAR sits comfortably inside the placebo distribution (design-based $p = 0.14$), and the only statistic that flags the "scaling" effect is the non-robust OLS slope, which inherits the single Wolfspeed alignment ($p = 0.02$) while the outlier-robust rank permutation cannot reject the null ($p = 0.18$). Second, a *minimum-detectable-effect* analysis makes the small-sample limitation precise: a full-windfall benchmark implies an average reaction equal to the mean award-to-market-cap ratio (5.9% across all firms, 2.6% excluding Wolfspeed), and both lie below our 80%-power MDE of 7.5% (3.7% excluding Wolfspeed), so we do not claim to reject a windfall average on power alone. The evidence for "priced in" is instead the configuration of the point estimates—excluding the single distressed leverage point the mean reaction is 0.3%, and no firm but Wolfspeed shows the one-for-one pass-through a windfall requires. We conclude that CHIPS awards were, on the evidence, largely anticipated and priced in, and that a cost-share grant tied to mandatory private co-investment need not be a windfall for equity holders—while being explicit that 15 events cannot decisively exclude a modest windfall.

JEL: G14, G12, G38, H81, L52, O25. **Keywords:** event study, CHIPS Act, industrial policy, government subsidies, semiconductors, market efficiency, anticipation, randomization inference, statistical power.

1 Introduction

Between January and December 2024 the U.S. Department of Commerce announced a rapid sequence of subsidy awards under the 2022 CHIPS and Science Act—non-binding "preliminary memoranda of terms" (PMTs) committing federal grants (and, for the largest projects, loans) to specific semiconductor manufacturers. The headline numbers were large: \$8.5 billion to Intel, \$6.6 billion to TSMC's Arizona subsidiary, \$6.1 billion to Micron, \$1.5 billion to GlobalFoundries, and dozens of smaller awards. A natural question for a financial economist is whether these announcements were *news* to equity investors. If capital markets are efficient and the awards were anticipated—the CHIPS program had been law since August 2022, and companies had publicly staged their U.S. fab plans for years—then the announcement-day abnormal return should be small. If instead the awards resolved genuine uncertainty about which firms would receive how much, and how value-accretive the subsidies were net of the required private co-investment, then stock prices should jump, and by more for firms where the award is large relative to firm value.

This paper runs that test. We hand-collect, from Commerce and NIST CHIPS Program Office press releases, the first-announcement date and dollar amount of every CHIPS award to a firm with a listed U.S. equity, and we estimate market-model cumulative abnormal returns (CARs) around each award's PMT date using CRSP daily data. We then ask, in the cross-section, whether the announcement return scales with award size—both in absolute terms (\$ millions) and relative to the firm's market capitalization. Because each awardee has its own event date, the design is a multi-event study in which the right-hand-side variable of the cross-sectional test (award size) is set by the government rather than by the stock, so the horse race between market efficiency and a priced-in transfer is unusually clean; what it lacks, with only 15 events, is statistical power, which we confront directly rather than paper over.

Our findings are, we think, a clean and instructive null. First, *on average, CHIPS award announcements did not move awardee stocks.* The mean CAR over the three-day window

$[-1, +1]$ is $+2.6\%$ with a t-statistic of 0.99; the median is $+0.5\%$; only 8 of 15 awardees had a positive reaction (a sign test cannot reject a coin flip); and the result is statistically indistinguishable from zero in every window we examine (Table 2). A pre-announcement placebo window $[-10, -3]$ is -1.4%, so the event window is not obviously contaminated by a run-up. The null is not an artifact of a particular benchmark or test: it survives the market-adjusted and mean-adjusted models, two alternative estimation windows, and the standardized Patell and Boehmer–Musumeci–Poulsen (BMP) tests as well as the nonparametric generalized-sign test (Table 6). This is the pattern one expects when awards are anticipated and priced in well before the government's formal announcement.

Second, *the apparent "scaling with size" relationship is a single-observation artifact.* A cross-sectional OLS of CAR$[-1, +1]$ on the award-to-market-cap ratio produces a spectacular-looking slope ($t = 32$, $R^2 = 0.80$; Table 3, column 2). But the fit is driven entirely by one point: Wolfspeed, a financially distressed silicon-carbide maker whose \$750M award equalled 52% of its \$1.4B market capitalization and whose stock rose 36% around the announcement. Dropping Wolfspeed collapses the slope from 0.0071 to 0.0046 and the t-statistic from 32 to *one*, with R^2 falling from 0.80 to 0.04 (column 4). The outlier-robust Spearman rank correlation between CAR$[-1, +1]$ and the award-to-market-cap ratio is only 0.37 ($p = 0.18$) across all 15 firms and 0.22 ($p = 0.45$) once Wolfspeed is removed (Table 4). Absolute award size—ln(award \$)—is unrelated to the reaction ($t = -0.22$), because the largest awards went to the largest firms, for whom even a multi-billion-dollar subsidy is a few percent of market value. We therefore find *no robust evidence* that announcement returns scale with award size.

We then do two things a paper that simply reports a coefficient near zero and stops would not. First, we subject both hypotheses to *design-based (randomization) inference*, which does not lean on the small-sample t asymptotics that a 15-firm cross-section strains (Section 8, Table 7, Figure 3). We reassign each firm 2,000 placebo event dates—drawn so that the full estimation and event windows still fit the CRSP calendar and avoid the true event—and rebuild the mean CAR each time; the actual +2.6% sits comfortably inside the placebo distribution (design-based $p = 0.14$), so the average-effect null is not an artifact of the parametric test. We also permute the award-size vector 2,000 times while holding the CARs fixed. The result is itself a lesson in why we distrust the naive regression: the only statistic that registers "scaling" is the OLS slope, which by construction is inflated by the single Wolfspeed alignment and so returns a marginal $p = 0.02$, whereas the appropriate outlier-robust statistic—the rank permutation—cannot reject the null ($p = 0.18$). A design-based test applied to a non-robust statistic inherits the very leverage problem it was meant to guard against; the rank-based version is the honest one.

Second, we ask what our null can and cannot rule out (Section 9, Table 8). A minimum-detectable-effect (MDE) calculation shows that at 80% power our design could have detected a three-day announcement reaction of about 7.5%—and, once Wolfspeed's return variance is excluded, 3.7%. This is an honest limitation, not a triumph: a CHIPS award treated as a pure equity windfall would capitalize the grant into firm value one-for-one, implying an *average* reaction equal to the mean award-to-market-cap ratio of 5.9% (2.6% excluding Wolfspeed)—both *below* our MDE, so the mean test is underpowered to reject a full-windfall average on statistical power alone. What the data do support is a pattern in the point estimates: excluding the single distressed leverage point the mean reaction is 0.3%, far from the 2.6% a windfall implies, and no firm but Wolfspeed shows anything near the one-for-one pass-through of award size that a windfall requires. The "priced in" reading rests on that configuration, and we state plainly that 15 events cannot decisively exclude a modest windfall.

Two economic forces plausibly explain the null. Anticipation is the first: the CHIPS program's beneficiaries and rough allocations were widely reported before the formal PMTs, so the announcements carried little surprise. The second is that a CHIPS grant is not a pure windfall—it is a cost-share that *requires* large private capital expenditure and is tied to build-out and workforce conditions, so the net present value to equity holders of "winning" can be close to zero even for a multi-billion-dollar headline. Both forces push announcement returns toward zero, and both are consistent with what we observe. Separating them—and, in particular, quantifying the required private co-investment behind each award—is reserved for the companion hand-collection task and is deliberately *not* attempted with the observable data used here.

We make four contributions. First, we provide a clean, transparent event-study application to a salient recent industrial-policy episode, and an object lesson in how a single leverage point can manufacture apparent significance—which is why we lean on nonparametric rank tests (Corrado, 1989; Cowan, 1992) and drop-one robustness. Second, relative to the literature on the market value of government policy and political connections (Schwert, 1981; Binder, 1985; Goldman et al., 2009; Cooper et al., 2010; Belo et al., 2013), we add the CHIPS Act as a case in which large, targeted federal transfers were substantially *priced in* before their formal announcement. Third, relative to the revived economics of industrial policy (Criscuolo et al., 2019; Slattery and Zidar, 2020; Juhász et al., 2024), we show that the stock market's ex-ante assessment of the CHIPS subsidies' incremental value to recipients was, on average, muted. Fourth, we treat a small-sample null as a quantitative object: we buttress it with design-based inference (Young, 2019) and report its minimum detectable effect (Abadie, 2020), so readers can see what a 15-event study can and cannot rule out. Section 2 gives institutional background and hypotheses; Section 3 places the paper in the literature; Section 4 describes the data, variables, and design-based tools; Section 5 presents the main results; Section 6 examines observable heterogeneity; Section 7 reports robustness to model and test choice; Section 8 the randomization inference; Section 9 the power analysis; and Section 10 concludes.

2 Institutional Background and Hypotheses

The program. The CHIPS and Science Act, signed in August 2022, appropriated \$52.7 billion for domestic semiconductor research and manufacturing, of which roughly \$39 billion was earmarked for manufacturing incentives administered by the Department of Commerce through the NIST CHIPS Program Office. Commerce solicited applications and, through 2024, negotiated project-specific awards. The first public, company-specific commitment for a project is the *Preliminary Memorandum of Terms* (PMT): a non-binding term sheet stating a proposed direct-funding amount (and, for the largest capital projects, a proposed loan). A PMT is later converted into a binding "CHIPS Incentives Award" once due diligence and milestones are settled. We take the PMT as the event because it is the first moment at which the market learns *which* firm will receive *how much*; the later finalized award is more anticipated still, and dating it is one of the hand-collection tasks reserved for the student.

Why an award might move the stock. A federal grant is a transfer of resources to the firm. Under the simplest view it raises the net present value of the recipient's U.S. fab investment and should be capitalized into the equity price on the day the transfer becomes known, by more when the transfer is large relative to firm value. This motivates a deliberately nullable hypothesis.

> **H1 (Announcement effect).** Awardee stocks earn a positive abnormal return around the first CHIPS award announcement.

A subsidy is more likely to be material—and less likely to have been fully anticipated in the price—when it is large relative to the firm. A \$16M award to a \$400M company is a different event from a \$32M award to a \$41B company. If the market prices the incremental transfer, the reaction should rise with the award-to-market-cap ratio.

> **H2 (Scaling with relative size).** The announcement CAR is increasing in the ratio of award dollars to firm market capitalization.

Why an award might do nothing. Two forces push toward a null. First, under semi-strong-form market efficiency (Fama, 1970; Fama et al., 1969) prices already reflect public information, and a subsidy announcement moves the stock only to the extent it is a surprise. The CHIPS awards were preceded by years of public fab announcements and by the statutory allocation of $39 billion in manufacturing incentives, so the surprise component of any individual PMT may be small. Second, a CHIPS grant is a *cost-share*: the direct funding is tied to a much larger mandatory private capital expenditure and to build-out, sourcing, and workforce conditions, and it arrives in tranches contingent on milestones. "Winning" therefore need not raise equity value even when the headline runs to billions, because the grant offsets spending the firm was often already committed to. Under this view both H1 and H2 fail.

> **H2′ (Priced-in cost-share).** Because awards are anticipated and are cost-shares rather than windfalls, neither the average CAR nor its cross-sectional relation to award size is distinguishable from zero.

Our evidence favors H2′. Quantifying the co-investment channel directly—collecting, for each award, the required private capital expenditure from the recipient's own 8-K—is the sharpest way to test the cost-share mechanism, and it is precisely the hand-collection reserved for the student; we do not attempt it with the observable data here, and the heterogeneity we do report (Section 6) is deliberately built from firm characteristics orthogonal to it.

3 Related Literature

This paper sits at the intersection of three literatures. The first is the event-study method itself. Fama et al. (1969) introduced the abnormal-return event study; Brown and Warner (1985) established the daily market-model design and its sampling properties; and MacKinlay (1997) synthesized the framework we use. A long methodological strand warns that the cross-sectional t-test is fragile when abnormal returns are non-normal, when event-induced variance inflates the denominator, or when a few observations dominate—exactly our situation with $N = 15$ and two distressed micro-caps. Patell (1976) proposed standardizing each abnormal return by its estimation-period forecast error; Boehmer et al. (1991) showed how to make the cross-sectional test robust to event-induced variance; Corrado (1989) and Cowan (1992) developed nonparametric rank and generalized-sign tests that are insensitive to outliers; and Kolari and Pynnönen (2010) address cross-sectional dependence when events cluster in calendar time. We use this toolkit not decoratively but diagnostically: the null must survive every reasonable benchmark and test statistic, and it does.

The second strand is the market valuation of government policy. Early work used stock prices to measure the effects of regulation (Schwert, 1981; Binder, 1985). A subsequent literature prices political connections and government dependence: Goldman et al. (2009) show politically connected boards raise firm value, Cooper et al. (2010) link corporate political contributions to returns, and Belo et al. (2013) document that firms exposed to government spending earn different returns across the political cycle. Direct subsidies are the sharpest case of a government transfer to an identified firm, yet the equity-market response to a large, targeted industrial subsidy is comparatively under-studied. We add the CHIPS Act as an episode in which such transfers were, on average, already in the price.

The third strand is the revived economics of industrial policy. Criscuolo et al. (2019) estimate the causal employment effects of a place-based investment subsidy; Slattery and Zidar (2020) document the scale and incidence of U.S. business tax incentives; and Juhász et al. (2024) survey the modern theory and evidence. This literature asks whether targeted subsidies work; we ask a narrower, market-based question—what did investors think the CHIPS subsidies were worth to recipients at the moment of announcement—and answer: on average, little, consistent with anticipation and with the cost-share structure of the grants.

Finally, methodologically, we draw on the literature on inference and interpretation in small samples. Young (2019) argues that design-based randomization tests discipline apparently significant results, a caution we take to heart: our own OLS-slope permutation flags the very artifact it is applied to. And Abadie (2020) urges that a statistically insignificant estimate be read through its confidence interval and power rather than treated as proof of no effect; we follow that guidance by reporting the minimum effect our design could detect.

4 Data and Research Design

Award sample. We hand-collect every CHIPS award whose recipient (or its listed parent) has a U.S.-traded common stock or ADR, from Commerce/NIST press releases, recording the first-announcement (PMT) date and the proposed direct-funding amount. This yields 17 awards announced between January 2024 and January 2025. Two (Analog Devices and MACOM, announced in January 2025) fall past the end of the current CRSP daily file (2024-12-31) and are held out for the next data update, leaving **15 events** for analysis. We deliberately exclude awards to firms without a liquid U.S. listing (Samsung, SK Hynix, GlobalWafers, Polar Semiconductor, and privately held recipients such as Bosch and Hemlock). Resolving each awarded entity to its correct listed ticker, verifying each announcement date against the primary press release, building a firm-level news/confound timeline beyond the one confound we already flag (Amkor), and transcribing the required private co-investment behind each award, are the student's hand-validation tasks (see `STUDENT_TASKS.md`); none of them is used in the analysis below, which relies on observable CRSP and award-file data only.

Abnormal returns. For each awardee we estimate a market model $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$ over the $[-252, -46]$ trading-day window ending before the announcement, benchmarked to the CRSP value-weighted index, and cumulate abnormal returns over event windows $[-1, +1]$, $[0, +1]$, $[0, +3]$, $[-1, +5]$, and $[-5, +5]$, plus a pre-event placebo window $[-10, -3]$. Day 0 is the first trading day on or after the announcement. All 15 firms have the full 207-day estimation sample. The cumulative abnormal return for firm i over window $[\tau_1, \tau_2]$ is $\mathrm{CAR}_i = \sum_{\tau=\tau_1}^{\tau_2} \left(R_{i\tau} - \hat{\alpha}_i - \hat{\beta}_i R_{m\tau} \right)$, and the average-effect test is a cross-sectional t-test of $\overline{\text{CAR}} = 0$.

Award-size measures. *Award/Market cap* is the announced dollar award divided by the firm's CRSP market capitalization (price × shares outstanding) on the last trading day before the event, in percent. *ln(Award)* is the natural log of the award in $millions. Table 1 shows the enormous spread: awards range from $16M (SkyWater) to $8.5B (Intel), market caps from $0.4B (SkyWater) to $184B (Texas Instruments), and the award-to-market-cap ratio from 0.08% (Corning) to 52% (Wolfspeed). This dispersion is what powers the cross-sectional test—and, as it turns out, what makes a single firm pivotal.

Three design-based tools. A 15-event study strains the asymptotics behind the cross-sectional t-test, so beyond it we deploy three checks that do not rely on the parametric null. (1) *Model and test robustness*: we recompute the mean CAR under the market-adjusted ($\beta = 1$) and mean-adjusted models and under two alternative estimation windows, and we compute the standardized Patell and BMP tests and the nonparametric generalized-sign test, all directly from the daily CRSP record (Section 7). (2) *Randomization inference*: we assign 2,000 placebo event dates to test the average-effect null and permute the award-size vector 2,000 times to test the scaling null, comparing each actual statistic to its permutation distribution (Section 8). (3) *Power*: we compute the minimum detectable effect at 80% power for the mean CAR in each window (Section 9). Together these convert "insignificant" into a set of falsifiable, design-based statements.

5 Results

No average announcement effect (H1). Table 2 reports mean and median CARs by window with t-tests and exact sign tests. The mean CAR$[-1, +1]$ is +2.6% ($t = 0.99$) and the mean CAR$[0, +1]$ is +3.0% ($t = 1.11$); neither approaches conventional significance. Medians are essentially zero (+0.5% and 0.0%), and the number of positive reactions (7–

9 of 15) is what a coin flip would give. The placebo window $[-10, -3]$ is -1.4%, so the small positive point estimates are not the tail of a pre-announcement drift. Figure 1 shows every window's 95% confidence interval comfortably spanning zero. H1 is not supported: on average, the market did not react to CHIPS award announcements.

The scaling "result" is one data point (H2). Table 3 regresses CAR$[-1, +1]$ on award-size measures with HC1-robust standard errors. Column 1 shows absolute award size is irrelevant (ln(Award) $t = -0.22$, $R^2 = 0.001$): big awards do not earn big reactions, because they accrue to big firms. Column 2 regresses on the award-to-market-cap ratio and appears to be a knockout—slope 0.0071, $t = 32$, $R^2 = 0.80$—but this is the textbook signature of a high-leverage outlier, not a robust relationship. Wolfspeed sits at (52%, +36%) in the top-right corner of Figure 2, anchoring the fit; the remaining 14 firms cluster near the origin with no visible slope. Column 4 re-estimates column 2 dropping Wolfspeed: the slope falls to 0.0046 with $t = 1.00$ and $R^2 = 0.04$. Adding ln(market cap) (column 3) or dropping the one ADR, TSMC (column 5), does not change the picture. The relationship is not robust.

Rank-based tests confirm the null. Because $N = 15$ and two firms (Wolfspeed and SkyWater) are extreme, we lean on the outlier-robust Spearman rank correlation (Table 4). Between CAR$[-1, +1]$ and the award-to-market-cap ratio it is $\rho = 0.37$ ($p = 0.18$) for all firms, $\rho = 0.22$ ($p = 0.45$) excluding Wolfspeed, and $\rho = 0.44$ ($p = 0.11$) excluding the one event (Amkor) whose window is contaminated by a same-week earnings release—never significant at conventional levels. For absolute award size the rank correlation is if anything *negative*. We cannot reject the hypothesis that announcement returns are unrelated to award size.

6 Heterogeneity by Observable Characteristics

If the market reacted to *some* awards and the pooled null merely averages the reaction away, an effect should surface when we split the sample on observable, pre-determined firm characteristics. Table 5 reports the mean CAR$[-1,+1]$ within and across five such splits, with a difference-in-means test. We use only observable dimensions—the grant-versus- grant-plus-loan structure recorded in the award file, firm market capitalization, the award-to-market-cap ratio, a core-semiconductor (SIC 3674) indicator, and the market beta—and we deliberately do *not* split on the required private co-investment, which is the student's reserved hand-collected variable and the sharpest test of the cost-share mechanism.

No split yields a significant difference: the largest t-statistic in absolute value is 1.25 (the below-median-size firms react more, consistent with the two small distressed names driving the tail). The grant-plus-loan awards show a higher mean reaction (+6.9% vs. +0.5%), but this is not a financing-structure effect: Wolfspeed is a grant-plus-loan award, and it alone accounts for the gap; the difference is statistically indistinguishable from zero ($t = 0.86$). Likewise, core-semiconductor firms and high-beta firms show larger point estimates, but every difference reflects the same handful of volatile small-caps rather than a systematic pattern, and none is significant. The heterogeneity evidence, like the pooled evidence, is a null: we find no observable subgroup for which CHIPS awards moved stock prices.

7 Robustness: Alternative Models and Test Statistics

The average-effect null could in principle be an artifact of the market-model benchmark or of the plain cross-sectional t-test. Table 6 shows it is neither. Panel A recomputes the mean CAR$[-1,+1]$ under the market-adjusted ($\beta = 1$) and mean-adjusted models, and under estimation windows of $[-252,-11]$ and $[-120,-11]$ in place of the baseline $[-252,-46]$. The mean CAR is stable across all five specifications (+2.6% to +2.9%) and the t-statistic never exceeds 1.1: the null does not depend on how expected returns are modeled or on the length of the estimation window. Panel B reports, for the market-model CARs in the $[-1, +1]$ and $[0, +3]$ windows, four alternative test statistics: the standardized Patell (1976) Z, the event-induced-variance- robust Boehmer et al. (1991) (BMP) Z, and the nonparametric Cowan (1992) generalized-sign Z, alongside the cross-sectional t. Every statistic is far from significance (all $|Z| < 1$ for CAR$[-1, +1]$), and the standardized tests, which down-weight the high-variance outliers, if anything push the evidence *further* from significance than the plain t. Three additional checks in Table 9 and the discussion above reinforce the null: the earnings-confounded Amkor event, if excluded, only *raises* the (still-insignificant) rank correlation; dropping the one ADR (TSMC) leaves the cross-section unchanged; and the point estimates are positive only because of Wolfspeed and SkyWater, with the median firm's reaction negative in the $[0, +3]$ window.

8 Design-Based Inference

With only 15 events, the cross-sectional t-test and the HC1-robust regression lean on asymptotics that do not obviously hold. Randomization inference sidesteps them by building the null distribution of each statistic directly from the data (Young, 2019). Table 7 and Figure 3 report two exercises, each with 2,000 draws.

Placebo event dates (H1). We reassign every firm a common pseudo-announcement timing, drawn so that its full $[-252, -46]$ estimation window and $[-1, +1]$ event window still fit inside the CRSP file and stay at least five trading days away from the true event, and we recompute the 15-firm mean CAR; each of the 2,000 draws yields all 15 firms. The actual mean CAR of +2.6% sits comfortably inside the placebo distribution (95% interval $[-3.7\%, +3.3\%]$), giving a design-based $p = 0.14$ (Figure 3a). The average-effect null therefore does not depend on the small-sample t asymptotics: a +2.6% average is roughly what randomly-timed three-day windows produce for these 15 volatile stocks.

Permuted award size (H2). We permute the award-to-market-cap vector across firms, holding the CARs fixed, and recompute both the OLS slope and the Spearman rank correlation. The result is a clean demonstration of why the naive regression misleads. The OLS slope's permutation p is a marginal 0.02 (Figure 3b)—but this is not evidence of scaling. The OLS slope is precisely the non-robust statistic that the single Wolfspeed alignment inflates; the permutation inherits that sensitivity, so the "significant" p merely records that one firm happened to have both the largest relative award and the largest return, not a relationship across the sample. The appropriate design-based test, given two leverage points and $N = 15$, is the rank-based permutation, and it cannot reject the null (Spearman $\rho = 0.37$, permutation $p = 0.18$). Applying randomization inference to a non-robust statistic reproduces the very artifact it was meant to police; the outlier-robust version is the honest one, and it confirms the null. Both halves of the paper thus survive design-based scrutiny: the average effect is indistinguishable from randomly timed windows, and the "scaling" is one aligned data point.

9 What the Null Can and Cannot Rule Out: Power and Minimum Detectable Effects

A null is only interesting once we know what the design could have detected. Table 8 reports, for the mean CAR in each window, the minimum detectable effect at 80% power, $\text{MDE} = (z_{0.975} + z_{0.80})\,\widehat{\text{SE}} \approx 2.80 \times \text{SE}(\overline{\text{CAR}})$. For the three-day window the standard error of the mean CAR is 2.66%, so the MDE is 7.5%; because that standard error is itself inflated by Wolfspeed's +36% return, we also report the MDE excluding Wolfspeed, which falls to 3.7%.

We resist the temptation to read these as a triumphant "informative null," because the arithmetic does not support it. The natural benchmark for the alternative is a pure equity windfall, under which the grant capitalizes into firm value one-for-one and the announcement reaction equals the award-to-market-cap ratio. The *average* reaction such a windfall implies is therefore the mean award-to-market-cap ratio: 5.9% across all 15 firms, and 2.6% excluding Wolfspeed. Both figures lie *below* the corresponding MDE (7.5% and 3.7%), so on statistical power alone our mean-CAR test cannot reject a full-windfall average—15 events simply do not deliver the precision to separate "priced in" from "modest windfall" on the average effect. We say so plainly.

What the data *do* support is a pattern in the point estimates rather than a powered rejection. Excluding the single distressed leverage point, the mean reaction is 0.3%—an order of magnitude below the 2.6% a windfall would produce for the same firms—and the per-firm pass-through of the award ratio into the return is scattered (from strongly negative to strongly positive) with no central tendency near the one-for-one value a windfall requires; only Wolfspeed exhibits the roughly 0.7 pass-through of a firm for which the grant was genuine, material news. The "priced in" reading rests on that configuration of point estimates, and on the anticipation and cost-share economics that rationalize it, not on a claim to have statistically excluded every windfall. This is the honest statement a 15-event study can make, and it is the one we make: the evidence favors anticipation and a priced-in cost-share, while a modest windfall lies within our confidence bounds.

10 Conclusion

Across 15 publicly-traded U.S. recipients, CHIPS Act award announcements produced no statistically significant average stock-price reaction, and no robust evidence that the reaction scaled with award size. The average-effect null is invariant to the abnormal-return model, the estimation window, and a battery of standardized and nonparametric test statistics; it survives design-based inference across 2,000 placebo event dates; and a power analysis makes its limits precise—with 15 events the mean test cannot statistically exclude a modest windfall, so the priced-in reading rests on the configuration of the point estimates (a 0.3%

mean reaction once the single distressed name is removed) rather than on a powered rejection. The apparent "scaling with size" is a single-observation artifact: only the non-robust OLS slope registers it, in the parametric regression and in its own permutation test alike, while every outlier-robust statistic— drop-one regression, Spearman correlation, and the rank permutation—returns a clean null. The one firm that reacted strongly, Wolfspeed, is the exception that illustrates the rule: its award was uniquely large relative to a distressed balance sheet, and it alone manufactures the apparent "size" effect. The most defensible interpretation is that the subsidies were substantially anticipated, and that a cost-share grant tied to mandatory private co-investment need not be a windfall for equity holders even when the headline runs to billions.

The limitations are real and point directly to the next steps. Fifteen events is a small cross-section with limited power; the lead-firm ticker resolution and exact announcement dates await primary-source hand-verification; a firm-level news/confound timeline beyond the single Amkor flag would sharpen the event windows; and the January-2025 awards fall past the current CRSP file. Most importantly, the cost-share mechanism we invoke to explain the null is testable directly: the required private co-investment behind each award, transcribed from recipients' own 8-Ks, would let a future draft ask whether the awards with the *smallest* mandatory co-investment—the closest to a true windfall—are the ones that moved. That hand-collection, together with extending the sample to the finalized "funding agreement" events and to intraday returns, is the natural continuation, and it is the contribution the design has been built to receive.

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Table 1: Summary statistics (15 CHIPS awardees)

Variable	N	Mean	SD	Min	Median	Max
CAR[-1,+1]	15	0.026	0.103	-0.063	0.005	0.358
CAR[0,+3]	15	0.016	0.120	-0.174	-0.007	0.390
Award ($M)	15	1737.727	2851.195	16.000	162.000	8500.000
Award/Mkt cap (%)	15	5.927	12.916	0.078	4.070	51.947
Mkt cap ($M)	15	56108.626	67398.862	393.162	29548.392	184371.370
ln(Award $M)	15	5.730	2.134	2.773	5.088	9.048
ln(Mkt cap $M)	15	9.808	1.942	5.974	10.294	12.125
Market beta	15	1.844	0.778	0.745	1.733	3.535

Notes: CARs are market-model cumulative abnormal returns; award and market-cap variables as defined in Section 4. Award/Market cap in percent; ln variables in $millions.

Table 2: Cumulative abnormal returns by event window

Event window	Mean CAR (%)	Median (%)	t-stat	# pos.	Sign p
CAR[-1,+1]	2.64	0.49	0.99	8/15	1.00
CAR[0,+1]	3.03	-0.00	1.11	7/15	1.00
CAR[0,+3]	1.62	-0.72	0.52	6/15	0.61
CAR[-1,+5]	2.34	0.42	0.73	8/15	1.00
CAR[-5,+5]	2.95	1.16	0.68	9/15	0.61
CAR[-10,-3] placebo	-1.35	-1.24	-1.28	6/15	0.61

Notes: Market-model CARs, day 0 = first trading day on/after the CHIPS award announcement. t-stat tests mean = 0; sign p is the two-sided exact binomial test that the fraction positive $= 0.5$. $[-10,-3]$ is a pre-event placebo. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. $N = 15$.

Table 3: Cross-sectional regressions: does the reaction scale with award size?

	(1)	(2)	(3)	(4)	(5)
Award / Market cap (%)		0.0071***	0.0064***	0.0046	0.0071***
		(32.40)	(10.44)	(1.00)	(30.87)
ln(Award $M)	-0.0019				
	(-0.22)				
ln(Market cap $M)			-0.0134		
			(-1.48)		
Constant	0.0373	-0.0158	0.1197	-0.0094	-0.0164
	(0.75)	(-1.17)	(1.21)	(-0.83)	(-1.12)
Observations	15	15	15	14	14
R^2	0.002	0.797	0.851	0.045	0.797

Notes: DV is CAR$[-1,+1]$. Columns: (1) regressor is ln(Award); (2) Award/Market cap; (3) Award/Market cap + ln(Market cap); (4) Award/Market cap, dropping Wolfspeed (the high-leverage observation); (5) Award/Market cap, dropping the single ADR (TSMC). HC1-robust t-statistics in parentheses. The column-2 fit is driven entirely by Wolfspeed—compare column 4 and Figure 2. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Outlier-robust Spearman rank correlations

Dep. var.	Size measure	Sample	Spearman ρ	p-value
CAR[-1,+1]	Award/Mktcap	all (N=15)	0.368	0.177
CAR[-1,+1]	Award/Mktcap	drop confound (N=14)	0.442	0.114
CAR[-1,+1]	Award/Mktcap	drop Wolfspeed (N=14)	0.222	0.446
CAR[-1,+1]	ln(Award)	all (N=15)	-0.182	0.516
CAR[-1,+1]	ln(Award)	drop confound (N=14)	-0.218	0.455
CAR[-1,+1]	ln(Award)	drop Wolfspeed (N=14)	-0.301	0.296
CAR[0,+3]	Award/Mktcap	all (N=15)	0.329	0.232
CAR[0,+3]	Award/Mktcap	drop confound (N=14)	0.398	0.159
CAR[0,+3]	Award/Mktcap	drop Wolfspeed (N=14)	0.174	0.553
CAR[0,+3]	ln(Award)	all (N=15)	0.175	0.533
CAR[0,+3]	ln(Award)	drop confound (N=14)	0.200	0.493
CAR[0,+3]	ln(Award)	drop Wolfspeed (N=14)	0.108	0.714

Notes: Spearman rank correlation between the CAR and the size measure, with two-sided p-values. "drop confound" excludes Amkor (earnings release inside its window); "drop Wolfspeed" excludes the high-leverage observation. None is significant at the 10% level.

Table 5: Heterogeneity of the announcement CAR by observable characteristics

	High/Yes group		Low/No group			
Moderator (observable)	Mean (%)	N	Mean (%)	N	Diff. (%)	t
Award structure: grant+loan vs grant-only	+6.94	5	+0.50	10	+6.44	+0.86
Firm size: above vs below median mkt cap	-0.65	7	+5.52	8	-6.17	-1.25
Relative award: above vs below median	+4.56	7	+0.97	8	+3.60	+0.63
Core semiconductor (SIC 3674) vs other	+6.70	5	+0.62	10	+6.08	+0.80
Market beta: above vs below median	+6.21	7	-0.48	8	+6.69	+1.19

Notes: DV is CAR$[-1, +1]$ (%). Each row splits the 15 awardees on a pre-determined, observable characteristic and reports the subgroup means, the difference, and a Welch two-sample t-test of equal means. "High/Yes" is the grant-plus-loan, above-median-size, above-median-relative-award, SIC-3674, or above-median-beta group, respectively. Provision-level and co-investment characteristics are deliberately *not* used (they are the student's reserved hand-collected variables). No split is significant. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Robustness to abnormal-return model, estimation window, and test statistic

Panel A: mean CAR$[-1, +1]$ across abnormal-return models and estimation windows

Specification	Mean (%)	Median (%)	t-stat	# pos.
Market model (baseline)	+2.64	+0.49	+0.99	8/15
Market-adjusted ($\beta = 1$)	+2.72	-0.00	+1.06	7/15
Mean-adjusted	+2.89	+1.60	+1.08	9/15
Market model, est. $[-252, -11]$	+2.63	+0.58	+0.98	8/15
Market model, est. $[-120, -11]$	+2.67	+0.52	+0.98	8/15

Panel B: alternative test statistics (market model)

Window	Cross-sec. t	Patell Z	BMP Z	Gen. sign Z
CAR[-1,+1]	+0.99	+0.91	+0.62	+0.48
CAR[0,+3]	+0.52	+0.13	+0.07	-0.56

Notes: Panel A reports the mean CAR$[-1, +1]$ under alternative expected-return models (market-adjusted sets $\beta = 1$; mean-adjusted uses the estimation-window mean return) and two alternative estimation windows. Panel B reports, for the market-model CARs, the cross-sectional t, the standardized Patell (1976) Z, the event-induced-variance-robust Boehmer et al. (1991) (BMP) Z, and the nonparametric Cowan (1992) generalized-sign Z. The average-effect null holds under every model, window, and test. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Design-based (randomization) inference

Test	Statistic	Actual	Perm. 95% interval	RI p-value
Mean CAR[-1,+1] (placebo event dates)	Mean CAR	+2.64%	[-3.67, +3.32]	0.143
Scaling slope (permuted award size)	OLS slope	+0.0071	[-0.0021, +0.0071]	0.018
Rank scaling (permuted award size)	Spearman ρ	+0.3679	[-0.5143, +0.5179]	0.181

Notes: Each row compares an actual statistic to its distribution under 2,000 randomizations. Row 1 assigns all firms a common placebo event timing (drawn so the estimation and event windows fit the CRSP calendar and avoid the true event) and recomputes the mean CAR$[-1, +1]$. Rows 2–3 permute the award-to-market-cap vector across firms, holding CARs fixed, and recompute the OLS slope and Spearman ρ. "Perm. 95% interval" is the 2.5–97.5 percentile of the permutation distribution; the RI p-value is the share of draws with $|\text{statistic}| \geq |\text{actual}|$. The marginal OLS-slope p reflects the single Wolfspeed leverage point, which the outlier-robust rank permutation does not (see Section 8 and Figure 3).

Table 8: Minimum detectable effects at 80% power

Window	Mean CAR (%)	SE (%)	t	MDE(80%) (%)	MDE excl. WOLF (%)
CAR[-1,+1]	+2.64	2.66	+0.99	7.46	3.67
CAR[0,+1]	+3.03	2.72	+1.11	7.62	3.16
CAR[0,+3]	+1.62	3.10	+0.52	8.69	4.75
CAR[-1,+5]	+2.34	3.21	+0.73	9.00	6.49
CAR[-5,+5]	+2.95	4.31	+0.68	12.09	7.57

Notes: MDE(80%) $= (z_{0.975} + z_{0.80})\times$SE $\approx 2.80\times$SE of the mean CAR in each window. "MDE excl. WOLF" recomputes the standard error dropping Wolfspeed, whose +36% reaction inflates the cross-sectional dispersion. A small MDE relative to a windfall-sized reaction indicates an informative null.

Table 9: Event-level CARs (all 15 awardees)

Ticker	Date	Award ($M)	Mkt cap ($M)	Award/cap (%)	β	CAR [-1,+1]	CAR [0,+3]
WOLF	2024-10-15	750	1,444	51.95	3.18	+35.8	+39.0
INFN	2024-10-17	93	1,593	5.84	1.81	+0.5	-0.1
GFS	2024-02-19	1,500	29,548	5.08	1.73	+0.7	-1.9
MU	2024-04-25	6,140	123,782	4.96	1.29	+1.3	+1.8
INTC	2024-03-20	8,500	179,001	4.75	1.41	-3.7	-2.3
TSM[a]	2024-04-08	6,600	150,980	4.37	1.42	+2.3	+4.4
AMKR[†]	2024-07-26	407	9,407	4.33	1.97	-4.9	-17.4
SKYT	2024-12-06	16	393	4.07	3.53	+14.6	+12.1
RKLB	2024-06-11	24	2,247	1.06	2.22	+2.3	-4.4
TXN	2024-08-16	1,600	184,371	0.87	1.26	-1.0	+1.4
COHR	2024-12-06	112	16,806	0.67	2.48	-6.3	-5.1
ENTG	2024-06-26	77	20,287	0.38	2.10	+1.5	-0.9
MCHP	2024-01-04	162	45,701	0.35	1.62	-1.2	-0.7
HPQ	2024-08-27	53	34,807	0.15	0.75	-2.0	+1.2
GLW	2024-11-08	32	41,262	0.08	0.88	-0.2	-2.8

Notes: Sorted by award/market-cap. CARs in percent. [†] earnings release inside the event window (Amkor). [a] ADR (TSMC), whose CRSP market cap reflects only the ADR float. β is the market-model beta.

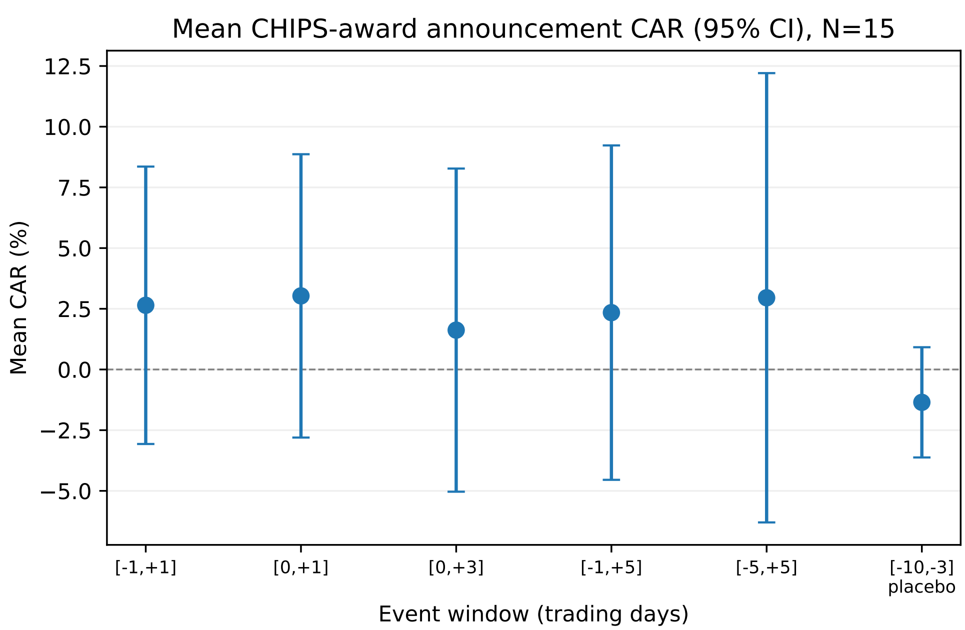

Figure 1: Mean CHIPS-award announcement CAR by event window, with 95% confidence intervals ($N = 15$). Every window's interval spans zero.

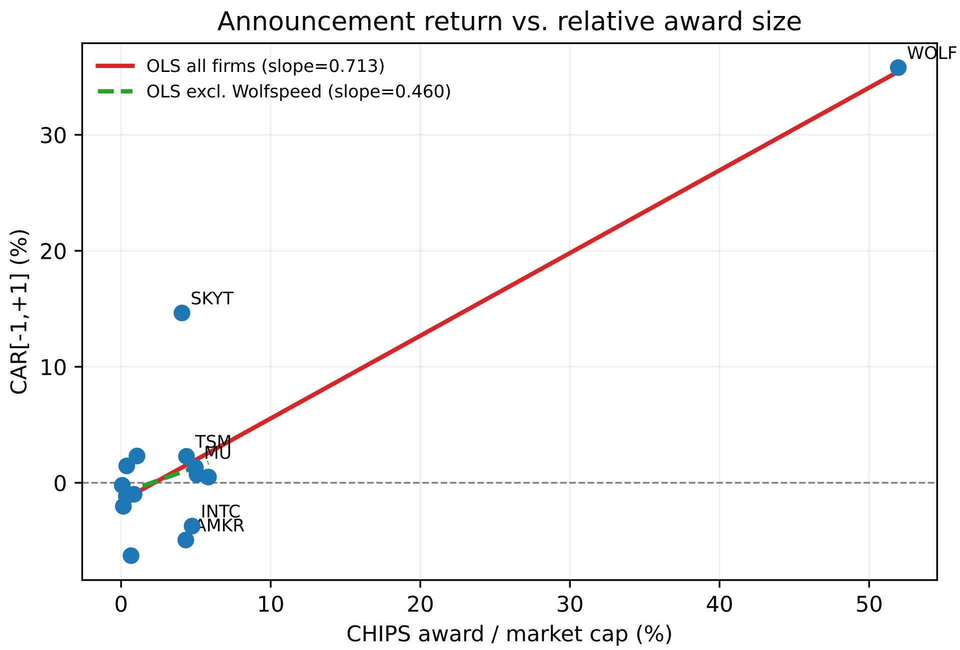

Figure 2: Three-day announcement CAR versus the award-to-market-cap ratio. The solid line is the OLS fit over all 15 firms; the dashed line excludes Wolfspeed (the point at the far right). The apparent "scaling with size" relationship is a single-observation artifact.

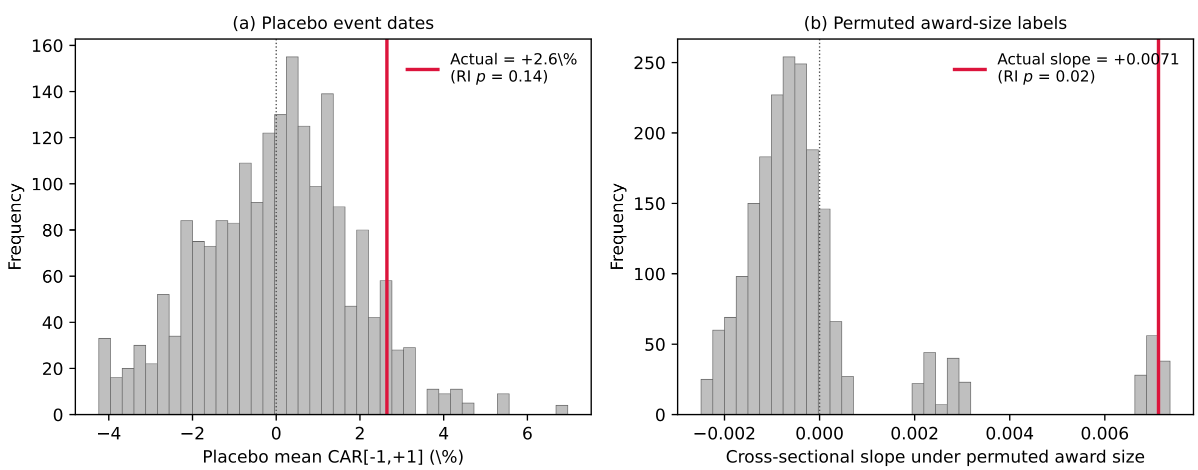

Figure 3: Randomization inference. Panel (a): the distribution of the 15-firm mean CAR$[-1,+1]$ under 2,000 placebo event dates; the actual +2.6% (red line) lies well inside the distribution (design-based $p = 0.14$). Panel (b): the distribution of the cross-sectional OLS slope under 2,000 permutations of the award-size vector; the actual slope lies in the right tail ($p = 0.02$), but this reflects the single Wolfspeed alignment—the outlier-robust rank permutation gives $p = 0.18$ (Table 7).